

Project 2

Geometry and Cuspidality in Orthogonal 3R Robots

Format: Group of two students; simulation-based project.

Prerequisites: D–H models, position IK, Jacobians, singularities, and configuration-space connectivity.

Motivation

Two robots with the same number of joints can have very different global inverse kinematics. A change in an axis offset can alter singularity curves, the distribution of inverse-kinematic solutions, and the possibility of changing posture without becoming singular. This project connects geometric design choices to the joint-space and workspace maps needed for motion planning.

Task

Overall objectives. Analyze a bounded family of orthogonal 3R position manipulators and explain how its geometry controls singularities, two-/four-IK regions, and evidence of cuspidal or noncuspidal behavior.

- **Define and validate the family.** Use the idealized family $a = (a_1, 2, 1.5)$, $d = (1, 1, 0)$, $\alpha = (\pi/2, \pi/2, 0)$, with $a_1 \in \{0, 0.5, 1\}$. State units and derive forward position kinematics. Check representative configurations using an independent implementation of the stated transforms. The zero-offset endpoint is a limiting geometry of this family, not the supplied custom 3R reference robot.
- **Construct reduced singularity plots.** Derive the position Jacobian J_p and check it by finite differences. Establish why $\det J_p$ is independent of q_1 . Plot its value and zero contours over $(q_2, q_3) \in [-\pi, \pi]^2$; identify sampled connected regions without equating equal determinant signs with connectivity.
- **Map the inverse distribution.** Map the singular curves to (ρ, z) , where $\rho = \sqrt{x^2 + y^2}$. Evaluate the supplied 3R IK method on a workspace grid and distinguish zero, two, and four distinct real solutions. Overlay the IK points of selected targets on the joint-space plot, including targets on opposite sides of a critical curve.
- **Investigate posture connectivity.** For each geometry, select one target with multiple regular IKs and examine whether two lie in the same computed singularity-free component. Seek one nonsingular connecting path where supported by the map. Explain what constitutes evidence of cuspidality and why failure to find a path does not prove noncuspidality.
- **Compare geometry, not plot appearance.** Summarize changes in IK counts, critical curves, and component membership. Refine the grids near a narrow region or suspected cusp. Support any definitive classification with an analytical argument and separate it from sampled observations; a numerical study alone need not provide a definitive classification.

Specifics and scope

Use exactly three geometries and at least three diagnostic workspace targets per geometry. Begin with approximately 150×150 joint and workspace grids and refine one region of interest. Treat joint angles as periodic for the unconstrained analysis. A complete symbolic classification of all orthogonal 3R geometries and a new quartic solver are not required.

Use the position Jacobian J_p for the 3R task. Confirm that the meridian reduction is valid for the selected model and operational point. Count distinct real IKs modulo periodicity, then filter joint limits separately. Identify two-/four-IK regions where present and explain any absent count rather than forcing both types into every map. Show both determinant values and the zero contour, and distinguish sampled connectivity from a proof. Refine the decisive region instead of claiming completeness from a coarse grid.

Expected outcome

Explain how D–H geometry changes global IK structure. Deliver three paired joint/workspace maps, an IK-count comparison, and one detailed connectivity example or a justified explanation of why the selected pair is separated.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Provided materials and implementation

The custom 3R robot description (.urdf) and link meshes (.stl) will be provided to the students, together with an analytical 3R IK implementation compatible with the stated parameter family. The reference robot is orthogonal with non-intersecting consecutive axes. It illustrates axis geometry; its dimensions must not be substituted for those of the idealized family.

Use standard D–H transforms $T_{i-1,i} = R_z(q_i)T_z(d_i)T_x(a_i)R_x(\alpha_i)$, with zero joint-angle offsets for the idealized family. Implement the parameter study using these transforms and use a plotting or robotics library to inspect frames, display IKs and singularities, and animate the selected connection. Validate the supplied IK results against your forward model.